17

Applications of basis reduction

This chapter presents four applications of basis reduction: breaking certain cryptosystems and linear congruential pseudorandom generators, finding simultaneous Diophantine approximations, and a refutation of Mertens' conjecture. We can only give the basic ideas; technical details can be found in the references, provided in the notes. The first two sections assume familiarity with the basics of cryptography, as explained in Chapter 20.

17.1. Breaking knapsack-type cryptosystems

The **subset sum problem** seeks to answer the following.

$$\text{Given } a_1,\dots,a_n,s\in\mathbb{N}, \text{ are there } x_1,\dots,x_n\in\{0,1\} \text{ with } \textstyle\sum_{1\le i\le n} a_i x_i = s? \qquad (1)$$

For example, an instance is to ask whether there exist $x_1,\dots,x_6\in\{0,1\}$ such that $366x_1+385x_2+392x_3+401x_4+422x_5+437x_6=1215$.

This problem is $\mathcal{NP}$-complete, as is a slight generalization of it, called the **knapsack problem**. After Diffie & Hellman (1976) invented public key cryptography, Merkle & Hellman (1978) proposed a public key cryptosystem based on the subset sum problem. The computations in this system were much less voluminous than for other systems such as RSA (Section 20.2), and its higher throughput seemed to promise a bright future. Several other such systems were proposed, based on versions of the knapsack problem. But the roof fell in when Shamir (1984) broke the Merkle & Hellman system, and almost all subsequently proposed improved schemes have suffered the same fate. Basis reduction has played a major role in some of these cryptanalyses.

In the notation of Section 20.1, Alice publishes her public key $a_1,\dots,a_n$ for such a **knapsack cryptosystem**. When Bob wants to send her n bits $x_1,\dots,x_n$ secretly, he encodes them as $s=\sum_{1\le i\le n} a_i x_i$ and sends s. Decoding a general such problem is $\mathcal{NP}$-complete and hence infeasible, but the idea now is to use a special type of problem for which decoding is easy with some secret additional knowledge,

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but hopefully hard without the secret. These special subset sum problems start with "superincreasing" $b_1 \ll b_2 \ll \cdots \ll b_n$ as the summands; a trivial example is $b_i = 2^{i-1}$, where the solution $(x_1, \ldots, x_n)$ is just the binary representation of s. More generally, it is sufficient to have $b_i > \sum_{1 \le j < i} b_j$ for all i; the solution is then unique and easy to calculate. The "easiness" is then hidden by multiplying the b_i's with a random number c modulo another random number m to obtain the public a_i's. Alice's private key c, m allows her to multiply s by c^{-1} modulo m and then solve an easy subset sum problem. At first sight, the a_i look like a general subset sum problem, but the cryptanalysts' work then showed that this hiding does not work. Of course, the breaking of these schemes does *not* mean that large instances of an $\mathcal{NP}$-complete problem can be solved routinely; the "superincreasing" subset sum problem is just too special.

The connection between the subset sum problem and short vectors is given by the fact that a solution of (1) yields a short vector in the lattice $L \subseteq \mathbb{Z}^{n+1}$ generated by the rows $r_1, \ldots, r_{n+1} \in \mathbb{Z}^{n+1}$ of the matrix

$$\begin{pmatrix} 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_n \\ 0 & 0 & \cdots & 0 & s \end{pmatrix} \in \mathbb{Z}^{(n+1)\times(n+1)}.$$

To see this, let $(x_1, \ldots, x_n) \in \{0,1\}^n$ be a solution of (1). Then

$$v = \sum_{1 \le i \le n} x_i r_i + r_{n+1} = (x_1, \ldots, x_n, 0) \in L$$

is a vector with $\|v\|_2 \le \sqrt{n}$, which is very small since the a_i are typically very large numbers. The approach to breaking such a cryptosystem is to compute a reduced basis of L and to hope that the resulting short vector is (essentially) v. Of course, this does not work too well for general subset sum problems, but it does work for **low-density subset sums**, where the ratio $n/(\max_i \log_2 a_i)$ of information bits to transmitted bits is small. This number is about $1/n$ for Merkle & Hellman's original scheme, and then this attack is very successful. In the example, the density $6/\log_2 437 \approx 0.684$ is high.

For example, with $a_1, \ldots, a_6$ as in the beginning, we consider the lattice $L \subseteq \mathbb{Z}^7$ generated by the rows of the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & -366 \\ 0 & 1 & 0 & 0 & 0 & 0 & -385 \\ 0 & 0 & 1 & 0 & 0 & 0 & -392 \\ 0 & 0 & 0 & 1 & 0 & 0 & -401 \\ 0 & 0 & 0 & 0 & 1 & 0 & -422 \\ 0 & 0 & 0 & 0 & 0 & 1 & -437 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1215 \end{pmatrix} \in \mathbb{Z}^{7\times 7}.$$

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Algorithm 16.10 then computes the short vector $v = (0,0,1,1,1,0,0) \in L$, and indeed $1215 = 366 \cdot 0 + 385 \cdot 0 + 392 \cdot 1 + 401 \cdot 1 + 422 \cdot 1 + 437 \cdot 0$.

17.2. Pseudorandom numbers

The simplest version of a **linear congruential pseudorandom number generator** is given by three parameters $a, b, m \in \mathbb{N}$, and for any seed $x_0 \in \{0, \ldots, m-1\}$ successively computes the elements $x_i \in \{0, \ldots, m-1\}$ defined by $x_{i+1} \equiv ax_i + b \bmod m$ for $i \in \mathbb{N}$. Pseudorandom numbers are used in many situations; the `rand` function in most computer systems is based on some form of these generators. These work just fine for computer algebra applications in modular algorithms or in the factorization of polynomials or integers. In cryptography, one would like to use them for the generation of keys. Therefore it is necessary that the numbers x_i are not only (almost) uniformly distributed but that it is practically impossible or computationally hard to infer x_i from $x_0, \ldots, x_{i-1}$, and to break a pseudorandom number generator means exactly to do this with a reasonable amount of time.

Boyar (1989) broke the simple generator given above by noting that

$$x_{n+1} - x_n \equiv a(x_n - x_{n-1}) \bmod m, \quad x_{n+2} - x_{n+1} \equiv a(x_{n+1} - x_n) \bmod m$$

imply that m divides $(x_{n+1} - x_n)^2 - (x_n - x_{n-1})(x_{n+2} - x_{n+1})$, provided that a and m are coprime. This yields guesses for m, a, and b. She uses these guesses to predict the sequence. If a prediction ever turns out to be wrong, this leads to a new, improved guess. She proves that in polynomial-time one arrives at a permanently correct prediction.

Frieze, Håstad, Kannan, Lagarias & Shamir (1988) considered a generator, which they attribute to the 1981 edition of Knuth (1998), where not the whole of x_i, but only the top half of the bits of x_i are used. They broke this using basis reduction, assuming that the generator m, a, b is known, but not the seed x_0.

The methods of both Boyar and Frieze *et al.* apply to much more general generators than explained here, and all these generators are useless for cryptographic purposes.

17.3. Simultaneous Diophantine approximation

We are given real numbers $\alpha_1, \ldots, \alpha_n$ and want to find $p_1, \ldots, p_n, q \in \mathbb{Z}$ such that $|\alpha_i - (p_i/q)|$ is "small" for all i and q is not too large. We can simply take q to be a power of 10 and p_i an appropriate initial segment of the decimal expansion of α_i. However, we look for much better solutions, with q much smaller than (the inverse of) the approximation error.

For $n = 1$, this is the classical Diophantine approximation problem, discussed in Section 4.6: the "best" approximations can be computed with the Extended Euclidean Algorithm and yield $|\alpha_1 - p_1/q| < 5^{-1/2} q^{-2}$.

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For general n, Dirichlet (1842) showed that there are infinitely many approximations with $|\alpha_i - p_i/q| \leq q^{-(1+1/n)}$ for all i. Lenstra, Lenstra & Lovász (1982) expressed this as a short vector problem, as follows. Given individual rational approximations $\alpha_i \approx \beta_i = u_i/v_i$ with $u_i, v_i \in \mathbb{Z}$ for $1 \leq i \leq n$ (but not necessarily the same denominator as demanded for a simultaneous approximation), and a rational ε with $0 < \varepsilon < 1$, we take $Q = \varepsilon^{-n}$ as an approximate bound on the denominator q. We set $w = 2^{-n(n+1)/4}\varepsilon^{n+1}$, and let $L \subseteq \mathbb{Q}^{n+1}$ be the lattice generated by the rows $f_0, \ldots, f_n \in \mathbb{Q}^{n+1}$ of the matrix

$$\begin{pmatrix} w & \beta_1 & \beta_2 & \cdots & \beta_n \\ 0 & -1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{pmatrix} \in \mathbb{Q}^{(n+1)\times(n+1)}.$$

In our treatment, we always assumed the vectors generating the lattice to have integral coefficients, but basis reduction also works for rational coefficients, as we have them here. It will produce in polynomial time a reduced basis for $L = \sum_{0\leq i\leq n} \mathbb{Z} f_i$. By multiplying together the $n+1$ inequalities in the proof of Theorem 16.9, we find that its first vector g satisfies

$$\|g\|^{2(n+1)} \leq \|f_0^*\|^2 \cdot 2\|f_1^*\|^2 \cdots 2^n \|f_n^*\|^2 = 2^{n(n+1)/2}\|f_0^*\|^2 \cdots \|f_n^*\|^2.$$

Since $f_0^*, \ldots, f_n^*$ are orthogonal, we find from Theorem 16.5 (iv)

$$\|g\| \leq 2^{n/4}(\|f_0^*\| \cdots \|f_n^*\|)^{1/(n+1)} = 2^{n/4} \left| \det \begin{pmatrix} f_0^* \\ \vdots \\ f_n^* \end{pmatrix} \right|^{1/(n+1)}$$

$$= 2^{n/4} \left| \det \begin{pmatrix} f_0 \\ \vdots \\ f_n \end{pmatrix} \right|^{1/(n+1)} = \varepsilon < 1.$$

Since $g \in L$, there exist $q, p_1, \ldots, p_n \in \mathbb{Z}$ so that

$$g = qf_0 + \sum_{1\leq i\leq n} p_i f_i = (qw, q\beta_1 - p_1, \ldots, q\beta_n - p_n).$$

If $q \leq 0$, we replace g by $-g$, so that $q \geq 0$, and $\|g\| < 1$ implies that $q \geq 1$. Then $|\beta_i - p_i/q| \leq \|g\|/q \leq \varepsilon/q = q^{-1}Q^{-1/n}$ for all i, and $1 \leq q \leq \varepsilon/w = 2^{n(n+1)/4}\varepsilon^{-n} = 2^{n(n+1)/4}Q$. This is about as good as guaranteed by Dirichlet's theorem, except for the factor $2^{n(n+1)/4}$; the latter does not depend on the sizes of the β_i, but only on their number.

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Lagarias (1985) gave the following all-integer variant. Input is the vector $\beta = (u_1/v_1, \ldots, u_n/v_n) \in \mathbb{Q}^n$. For any nonzero $q \in \mathbb{N}$, we write

$$\{\{\beta q\}\} = \min_{p_1,\ldots,p_n \in \mathbb{Z}} \max_{1 \le i \le n} \left\{ \left| \frac{u_i}{v_i} - \frac{p_i}{q} \right| \right\}$$

for the best approximation quality with denominator q. For p_i, one simply takes the integer nearest to $u_i q/v_i$. A further input is a bound Q on the denominator. If a simultaneous approximation denominator q^* exists with $1 \le q^* \le Q$ and $\varepsilon = \{\{\beta q^*\}\}$, say as guaranteed by Dirichlet's theorem but unknown, then the algorithm produces an approximation q which is almost as good:

$$1 \le q \le 2^{n/2} QV, \text{ and } \{\{\beta q\}\} \le \sqrt{5n}\, 2^{(n-1)/2} \varepsilon. \tag{2}$$

We let $V = v_1 \cdots v_n$, and assume that $\varepsilon > 0$. For all $j \in \{0, \ldots, n + \log_2(QV)\}$, we consider the lattice $L_j \subseteq \mathbb{Z}^{n+1}$ spanned by the rows of the matrix

$$\begin{pmatrix} 2^j & QV\dfrac{u_1}{v_1} & QV\dfrac{u_2}{v_2} & & QV\dfrac{u_n}{v_n} \\ 0 & QV & 0 & \cdots & 0 \\ 0 & 0 & QV & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & QV \end{pmatrix} \in \mathbb{Z}^{(n+1)\times(n+1)}.$$

The lower n vectors have only one nonzero entry each. We run the basis reduction algorithm 16.10 on this basis, and let $x^{(j)} = (x_0^{(j)}, \ldots, x_n^{(j)}) \in \mathbb{Z}^{n+1}$ be the short vector returned. Lagarias shows that for some value of j, the denominator $q = x_0^{(j)}$ provides an approximation satisfying (2), and that the whole algorithm runs in polynomial time.

EXAMPLE 17.1. We try Lagarias' method on the binary logarithms of the musical intervals $2, 3/2, 4/3, 5/4, 6/5, 9/8$ from Section 4.8. Since $\log_2 2 = 1$ and $\log_2(3/2) + \log_2(4/3) = 1$, it suffices to find simultaneous Diophantine approximations for $\alpha_1 = \log_2(4/3) \approx 0.42$, $\alpha_2 = \log_2(5/4) \approx 0.32$, $\alpha_3 = \log_2(6/5) \approx 0.26$, and $\alpha_4 = \log_2(9/8) \approx 0.17$. We take as initial approximation u_i/v_i the decimal expansion of α_i rounded to two digits, so that we start with a simultaneous Diophantine approximation with common denominator $V = 100$ (instead of the product required above). Letting $Q = 1$ and $j = 0$, we obtain the lattice $L \subseteq \mathbb{Z}^5$ generated by the rows of the matrix

$$\begin{pmatrix} 1 & 42 & 32 & 26 & 17 \\ 0 & 100 & 0 & 0 & 0 \\ 0 & 0 & 100 & 0 & 0 \\ 0 & 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 0 & 100 \end{pmatrix}.$$

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Then Algorithm 16.10 computes the matrix

$$\begin{pmatrix} 12 & 4 & -16 & 12 & 4 \\ -19 & 2 & -8 & 6 & -23 \\ 1 & 42 & 32 & 26 & 17 \\ 16 & -28 & 12 & 16 & -28 \\ 22 & 24 & 4 & -28 & -26 \end{pmatrix},$$

whose rows form a reduced basis of L. Indeed, the leftmost entries of the first two rows yield our familiar common denominator 12 and the next best choice 19. $\diamond$

17.4. Disproof of Mertens' conjecture

We defined the **Möbius function** $\mu(n)$ for $n \in \mathbb{N}$ in Section 14.9, equation (11). Its summation $M\colon \mathbb{N} \longrightarrow \mathbb{N}$ is given by $M(x) = \sum_{n \le x} \mu(n)$. Table 17.1 lists the first fifteen values of μ and M.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\mu(n)$	1	−1	−1	0	−1	1	−1	0	0	1	−1	0	−1	1	1
$M(n)$	1	0	−1	−1	−2	−1	−2	−2	−2	−1	−2	−2	−3	−2	−1

TABLE 17.1: The values of $\mu(n)$ and $M(n)$ for $n \le 15$.

Mertens (1897) contains a table and a foldout chart of values up to 10000, and he conjectured that $|M(x)| \le \sqrt{x}$ for all $x \in \mathbb{N}$; a similar conjecture had been made in 1885 by Stieltjes.

The conjecture may seem to come out of the blue, but in analytic number theory one studies various functions that take values $0, 1, -1$ (or other complex values with absolute value 0 or 1) such as the Jacobi symbol (Section 18.5) and proves that their sum up to x is absolutely bounded by $O(\sqrt{x})$. The same is true for the absolute value of the sum of a random sequence of 1 and -1 (Exercise 19.18; the sum itself has mean 0); in fact, the quotient $M(x)/x$ goes to zero if and only if μ takes the values 1 and -1 roughly equally often. This principle will motivate the bound on the size of elliptic curves (Hasse's theorem 19.20).

Mertens proved that his conjecture implies the famous Riemann Hypothesis (Notes 18.4). Furthermore, it was known that it implies the unsolvability of a certain (inhomogeneous) simultaneous Diophantine approximation problem, as in the previous section, involving roots of Riemann's zeta function. However, Odlyzko & te Riele (1985) used basis reduction in a lattice in $\mathbb{R}^{70}$ to show that this approximation problem does have a solution, and thus disproved Mertens' conjecture. Their account is eminently readable also for the non-specialist, and their method suggests that a counterexample might exist for an x of order $\exp(10^{65})$, but current algorithmics do not allow us to calculate M for such huge arguments. So we know that an x with $M(x) > 1.065\sqrt{x}$ exists, but we do not know any such x.

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Notes. 17.1. The subset sum problem was proven to be $\mathcal{NP}$-complete by Karp (1972); see problem SP13 of Garey & Johnson (1979). The original successful attack on the Merkle & Hellman system was by Shamir (1984). Lagarias & Odlyzko (1985) described the short vector attack, and Odlyzko (1990) gives a nice overview of the problem. Other subset sum cryptosystems were proposed by Graham & Shamir (see Shamir & Zippel (1980) for a description), Lu & Lee (1979), Niederreiter (1986), Goodman & McAuley (1984), Ong, Schnorr & Shamir (1984). Most of them were broken in the 1980s using basis reduction, among others by Adleman (1983) and Brickell (1984, 1985). The only knapsack type cryptosystem withstanding all attacks up to now is the Chor & Rivest (1988) scheme. Many further applications of basis reduction in cryptography are discussed in Nguyen & Stern (2001).

17.2. Lagarias (1990), §8, surveys pseudorandom number generators in cryptography.

17.3. Dirichlet (1842) showed that there exist simultaneous Diophantine approximations with absolute error bound $q^{-(1+1/n)}$, and Lagarias (1982a, 1982b) presents many results concerning best approximations. Lagarias (1985) discusses the computational complexity of various such problems, which, depending on the specification, ranges from polynomial-time to $\mathcal{NP}$-complete. As an example, the following is $\mathcal{NP}$-complete: given $\beta \in \mathbb{Q}^n$, as in Section 17.3, and integers Q, s, t, is there an approximation denominator q with $1 \leq q \leq Q$ and $\{\{\beta q\}\} \leq s/t$?

Exercises.

17.1$^{\longrightarrow}$ We consider the following knapsack cryptosystem. The pairs AA, AB, ..., AZ, BA, BB, ..., BZ, ..., ZA, ZB, ..., ZZ of letters are identified with the 10-bit representations of the numbers $0, \ldots, 26^2 - 1 = 675$. For example, the pair AL corresponds to the bit string $x_9 x_8 \cdots x_0 = 0000001011$. Longer messages are broken into two-letter blocks and each block is treated separately.

1. The private key is $c_9, \ldots, c_0, m, w \in \mathbb{N}$ with $c_{i+1} \geq 2c_i$ for $0 \leq i \leq 8$, $m > \sum_{0 \leq i \leq 9} c_i$, and $\gcd(w, m) = 1$.
2. The public key is $a_i = (wc_i \text{ rem } m) \in \mathbb{N}$ for $i = 0, \ldots, 9$.
3. A bit string $x = x_9 x_8 \cdots x_0$ is encrypted as $s = \sum_{0 \leq i \leq 9} x_i a_i \in \mathbb{N}$.
4. To decrypt a ciphertext s, you compute $t \in \mathbb{N}$ such that $t \equiv w^{-1} s \bmod m$ and $0 \leq t < m$. Then $t = \sum_{0 \leq i \leq 9} x_i c_i \in \mathbb{N}$, and you can reconstruct $x_9, x_8, \ldots, x_0$ from t.

(i) Write procedures for encryption and decryption, and check them with the key $c_0 = 1$ and $c_{i+1} = 2c_i + 1$ for $0 \leq i \leq 8$, $m = 9973$, and $w = 2001$, on the message "ALGEBRAISFUN".

(ii) Prove that $t = \sum_{0 \leq i \leq 9} x_i c_i$ actually holds in step 4.

(iii) Now you are an eavesdropper who knows the public key

i	9	8	7	6	5	4	3	2	1	0
a_i	2720	2580	5963	5712	7529	8393	6372	6749	6660	2775

and has intercepted a ciphertext consisting of the eight blocks

$$30\,754, 5\,495, 29\,503, 30\,781, 29\,048, 15\,752, 43\,586, 33\,361$$

(in decimal representation). Try basis reduction to find the original message. This need not work for all blocks.

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